

Section A

1	(a) region (of space) where a particle / body experiences a force	B1	[1]
	(b) similarity: e.g. force $\propto 1 / r^2$ potential $\propto 1 / r$	B1	[1]
	difference: e.g. gravitation force (always) attractive	B1	
	electric force attractive or repulsive	B1	[2]
	(c) <i>either</i> ratio is $Q_1 Q_2 / 4\pi\epsilon_0 m_1 m_2 G$	C1	
	$= (1.6 \times 10^{-19})^2 / 4\pi \times 8.85 \times 10^{-12} \times (1.67 \times 10^{-27})^2 \times 6.67 \times 10^{-11}$	C1	
	$= 1.2 \times 10^{36}$	A1	[3]
	or $F_E = 2.30 \times 10^{-28} \times R^{-2}$ (C1)		
	$F_G = 1.86 \times 10^{-64} \times R^{-2}$ (C1)		
	$F_E / F_G = 1.2 \times 10^{36}$ (A1)		

2 (a) amount of substance

M1